How-To Library

Informatica

Implement Disaster Recovery and High Availability for Enterprise Data Catalog

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Abstract

See the *Implement Disaster Recovery and High Availability for Enterprise Data Catalog* article for information about planning and implementing high availability and disaster recovery in Enterprise Data Catalog.

Supported Versions

- Informatica Enterprise Data Catalog 10.5

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Overview of High Availability in Enterprise Data Catalog

High availability eliminates a single point of failure in an Informatica domain and provides minimal service interruption in the event of failure. When you configure high availability for a domain, the domain can continue running despite temporary network, hardware, or service failures. You can configure high availability for the domain, application services, and application clients.

The following high availability components make services highly available in an Informatica domain:

- Resilience. An Informatica domain can tolerate temporary connection failures until either the resilience timeout expires, or until the failure is fixed.
- Restart and failover. A process can restart on the same node or on a backup node after the process becomes unavailable.
- Recovery. Operations can complete after a service is interrupted. After a service process restarts or fails over, it restores the service state and recovers operations.

The following Informatica platform services support automatic failover as part of high availability implementation in active-passive configuration:

- Domain gateway service
- Model Repository service
- Data Integration service
- Content Management service
- Catalog service

High availability features for the Informatica environment are available based on your license.

It is recommended that you have a multi-node domain installation. The infrastructure and profiling services must be installed on separate nodes. The Catalog Service must have a backup node configured.

High Availability Architecture

If your license includes high availability, you can configure Informatica platform services to run on multiple nodes. The nodes in the Informatica domain are enabled for high availability as long as there are at least three nodes in the cluster.

To avoid a single point of failure and have minimal service interruption in the event of a failure, the components of Enterprise Data Catalog, such as data storage, processing, and indexing services, run on each node. These nodes automatically implement service restart, failover, resiliency, and recovery. When the primary node is unavailable, the services can run on backup nodes. You can use an external load balancer to route data to the recovery nodes.

Enterprise Data Catalog stores the metadata extracted from external sources within the MongoDB database. To enable high availability, deploy MongoDB in active-active mode on multiple servers with a minimum requirement of three nodes.

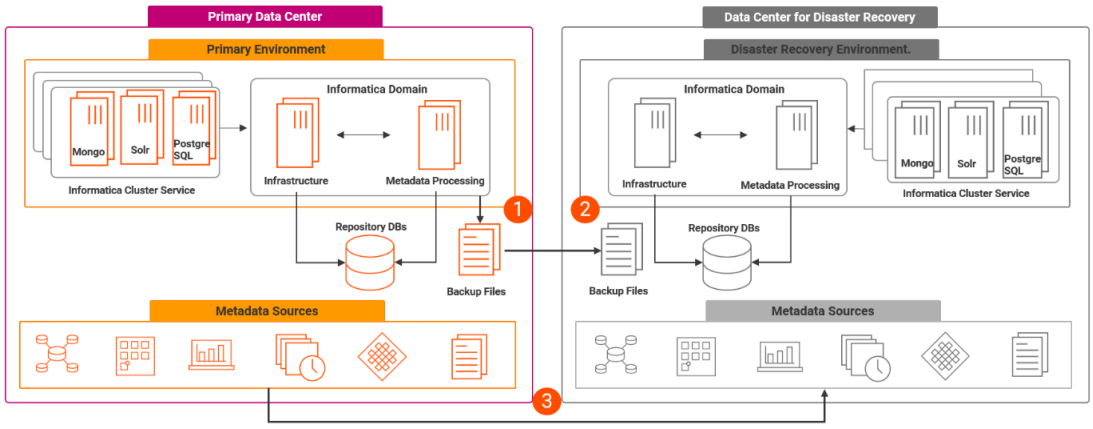
PostgreSQL is used as the store for similarity profiling data. The PostgreSQL database is deployed on a dedicated node and can only be scaled vertically. Therefore, the node does not support high availability. If a failure occurs, the functionality related to column similarity discovery is impacted and you cannot view similar columns in Enterprise Data Catalog. In this case, run the profile job on the resource again after you resolve the issue with the PostgreSQL database.

Overview of Disaster Recovery in Enterprise Data Catalog

Enterprise Data Catalog supports disaster recovery that helps your enterprise run the application services without a disruption in the event of a disaster. Disaster recovery includes configuring a recovery environment on the Informatica domain and replicating the application services, the Informatica domain, and the repositories between the production site and the recovery site. Follow the listed guidelines, workflow, and process to implement disaster recovery for Enterprise Data Catalog.

Disaster Recovery Architecture

The following image shows the disaster recovery architecture in Enterprise Data Catalog:



1. Regular backups are performed, and the backup files are copied to the disaster recovery environment.
2. During recovery, the backup is restored, and the Informatica Cluster service is started on the disaster recovery environment.
3. The dependent services and other applications are restored.

Disaster recovery Implementation Guidelines

Consider the following guidelines to implement disaster recovery in Enterprise Data Catalog:

- Set up multiple nodes in the production and recovery sites and set up the application services on these nodes. Configure primary nodes and their backup nodes.
- Register the nodes in the recovery site in the Informatica domain. After you configure the services on these nodes, stop the nodes.
- Make sure that the services are in a disabled state in the recovery site.
- Duplicate the primary Enterprise Data Catalog deployment to a disaster recovery environment in a different data center. The hosts in the disaster recovery environment must have the same hostnames as that of the corresponding hosts in the primary environment.
- Ensure that the dependent systems such as Active Directory and LDAP authentication systems are available in the disaster recovery environment.
- Backup the primary environment on a regular basis based on your SLA requirements. Stop the Catalog Service before you perform the backup.
- Back up the domain configuration, the Model Repository Service, and the reference database.
- Replicate the backup files to the disaster recovery environment.
- In the disaster recovery implementation, new instances of Analyst Service, Content Management Service, and Search Service are created on the recovery site. The services are configured to access a single Model Repository Service instance.
- You can replicate the profiling warehouse database on the recovery site for failover in the event of a production site crash.
- You can use virtual IP addresses when you configure databases at the recovery site.
- Ensure that you configure separate Informatica Cluster Services and Catalog Services on the production site and the recovery site.
- Back up the Catalog Service periodically.
- A new Catalog Service is configured on the recovery site to point to the same Model Repository Service instance accessed by the other services. Make sure that the Catalog Service is in a disabled state in the recovery site.
- Make sure that the Catalog Service is in the disabled state before you backup and restore the Catalog Service.
- Ensure that the configuration for the Catalog Service is the same on both the sites including the property file configuration.

Backup for Disaster Recovery Workflow

As part of disaster recovery, make sure that you perform the steps in the following order to back up the domain and the application services:

1. Back up the Enterprise Data Catalog using the `infacmd ldm BackupContents` command. The command creates a .zip file of the directory in the local machine:

   ```
   ./infacmd.sh ldm backupContents -dn <name of the Informatica domain> -un <username to connect to the Informatica domain> -pd <password for the username> -sn <name of the Catalog Service> -of <name of the backup ZIP file>
   ```

   For more information about the syntax, parameters, and description of the command, see Informatica Command Reference .

2. Back up the Informatica domain using the following command:

```
infasetup BackupDomain <-da database_hostname:database_port | -cs
database_connection_string> -du database_user_name -dp database_password -dt
database_type [-ds database_service_name] -bf backup_file_name [<-Force|-f>] -dn
domain_name
```

   See the Informatica Upgrade Guide for more information.

3. Back up the model repository using the following command:

```
infacmd mrs backupcontents
```

   Alternatively, you can use the informatica Administrator to back up the repository.

4. Back up the reference data location configured for the Content Management Service.

Disaster Recovery Workflow

In the event of a production site crash, restore the backups on the disaster recovery environment in the following sequence:

1. Restore the domain backup.
2. Restore the Model Repository Service backup.
3. Restore the reference database backup.
4. Restore the Catalog Service data using the CLI.

Update the DNS resolution to point to the host IP addresses from the disaster recovery environment. Review and validate the environment, and then enable the following application services:

1. Informatica Application Services
   - Model Repository Service
   - Data Integration Service
   - Content Management Service
2. Informatica Cluster Service
3. Catalog Service

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